A deterministic emulator of the Veros ACC last-year state

31 October 2026

Summary

- Goal: predict the last-year temperature of the Veros ACC channel directly from the EKE-closure coefficients (c_k, c_ϵ), skipping the 100-year integration.
- Method: a small deterministic network maps $z = (\log c_k, \log c_\epsilon)$, standardised, to the leading PCA coefficients of the training states; it is an ensemble of 5 initialisations. Hyper-parameters come from repeated cross-validation with the one-standard-error rule. A fit takes seconds on a CPU.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks, leave-one-out): RMSE 0.017–0.038 K.
  - The nearest training run gives 0.038–0.118 K.
  - The best classical emulator, a Gaussian process on one collapsed parameter s, gives 0.019–0.045 K. The network is lower on all five evaluations, by 0.0025–0.0075 K. With a Holm correction over the eight main evaluations, the difference is significant on four ($p \le 0.044$) but not on the scattered split ($p = 0.074$).
- Extrapolation (top row, outer 51, outer 75): 0.049 / 0.067 / 0.114 K.
  - The GP on s gives 0.079 / 0.074 / 0.125 K. The network is significantly better on outer 51 and outer 75 (Holm $p = 0.013$ and 0.004). On the top row most of the gain is one run, and the difference is not significant.
  - The nearest run gives 0.176 / 0.134 / 0.159 K.
  - The warm corner sets the error: 0.16 K one row from the training set, 1.06 K three cells away.
- The margin over the GP on s depends on the selection rule.
  - With the configuration of minimum CV error instead of the one-standard-error rule, the network is still lower on 7 of the 8 evaluations. It ties on the 7×7 block and on outer 75, and only leave-one-out stays significant after the Holm correction.
  - The one-standard-error rule was adopted after a first review had seen test errors, so its margins are not a blind test.
  - Against the GP on z, the RBF and the nearest run, the network is lower under both rules.
- Robustness: four initialisation sets move a split's mean by 0.0004–0.0010 K, and by 0.0028 K on the top row. Among the configurations that cross-validation cannot tell apart, the test error spans up to 0.03 K.
- Reproducibility: with one BLAS thread, a refit reproduces every number bit for bit. With several threads, a split's mean moves by up to 0.0003 K.
- With four training runs the hyper-parameters cannot be chosen. Depending on them, the network gets 0.10–0.19 K, against 0.120 K for the GP on s (Section 5).

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$ (ratio $2^{2/3}$); 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z) with a 930-cell ridge.
- Target: $y_r \in \mathbb{R}^{17970}$, the mean temperature of the last 12 outputs (360 days) of run r on the water cells. $S = 35$ psu everywhere, so it is dropped.

- The runs still drift, so the averaging window matters. Over the last 20 years, the RMS change from the first to the last year is 0.16 K, and the last-year mean differs from the 20-year mean by 0.08 K.
- The two parameters act through one combination, $s = \log c_k - 1.08 \log c_\epsilon$ (Figure 1). Below $s \approx -2$ (48 runs) the domain-mean T stays within 0.09 K; above, it rises by 1.1 K.

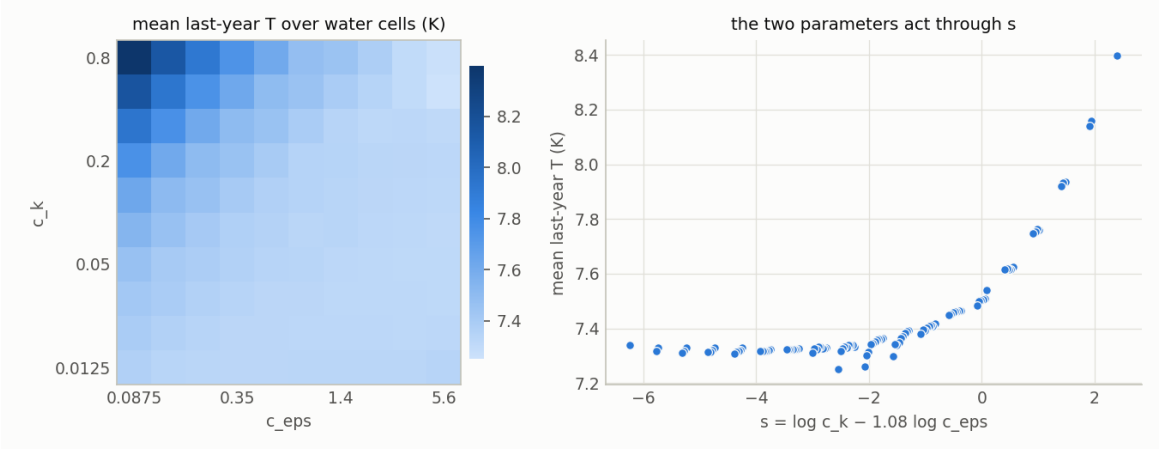


Figure 1: Mean last-year T over the water cells: over the parameter grid (left) and against s (right), the direction along which the first PCA coefficient of the states varies linearly.

2 Method

- Input z: $(\log c_k, \log c_\epsilon)$ standardised over the design.
- Output: a field in the span of the training states,

$$\hat{y}(z) = \mu + \sum_{k=1}^{K} a_k(z)\, v_k, \qquad K = \min(20,\, n-1), \tag{1}$$

  where μ and v_k are the mean and the leading principal directions of the n training states. So $K = 20$ on every split except the four-run one ($K = 3$).
- Network $a(z)$: a multilayer perceptron $2 \to 64 \to 64 \to K$ (scikit-learn, L-BFGS). Its targets are $a_k(z_r) = v_k^\top (y_r - \mu)$ divided by $\mathrm{std}(a_1)$, so with orthonormal v_k the data term is the physical MSE of the projected field up to a constant. In scikit-learn's normalisation:

$$\mathcal{L} = \frac{1}{2nK} \sum_{r=1}^{n} \frac{\|a(z_r) - V^\top (y_r - \mu)\|^2}{\mathrm{std}(a_1)^2} + \frac{\alpha}{2n} \|W\|^2. \tag{2}$$

- Ensemble: the prediction is the mean of 5 networks that differ only in their initialisation. The whole evaluation is repeated with 4 such sets, and reported values average them.
- Hyper-parameters: $(\text{activation}, \alpha) \in \{\mathrm{ReLU}, \tanh\} \times \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, chosen on the training runs of each split.
  - The search is 5-fold cross-validation repeated 3 times. A configuration's CV RMSE is its RMSE per training run, averaged over the repeats and then over the runs.
  - The one-standard-error rule then picks the most regularised configuration whose CV RMSE is within one standard error of the best. That standard error is over runs: the standard deviation of the best configuration's per-run CV RMSE, over $\sqrt{n}$. It is larger than a standard error over folds, so the rule leans towards regularisation.
  - The choice is ReLU with $\alpha = 10^{-2}$ on every split except outer 75 ($\alpha = 10^{-1}$) and four runs ($\alpha = 1$).
  - The minimum of the same scores is ReLU with $\alpha = 10^{-3}$ or 10^{-4}, except on outer 75 (10^{-2}) and four runs (1).

  - The rule is not blind to the test sets. It replaced a single 5-fold cross-validation that took the minimum, after an independent review had shown that choice to be decided by fold noise, and that review had already computed test errors of other configurations. Table 2 therefore also gives the network chosen by the minimum of the same cross-validation scores.
  - For leave-one-out the choice is made once, on all 100 runs. Each held-out run therefore takes part in it: a mild selection leak, with flat scores.
- Fitting on run means rather than on snapshots changes nothing: $\sum_t \|f(z_r) - x_{r,t}\|^2 = n_t \|f(z_r) - \bar{x}_r\|^2 + \text{const}$.
- Baselines, none tuned by cross-validation (unlike the network):
  - the training mean;
  - the nearest training run in z;
  - thin-plate RBF interpolation of every cell (exact, no smoothing);
  - a GP per PCA coefficient on z (Matérn 5/2, one length scale per input, white noise up to the full variance; hyper-parameters by maximum marginal likelihood);
  - the same GP on $s = w^\top z$, with w from a linear fit of a_1 on z.

3 Evaluation

- Eight hold-out splits of the grid, plus leave-one-out over the 100 runs:
  - interpolation: 10 scattered interior runs; the rows $c_k = 0.126$, 0.2, 0.3175; the centre 5×5 and 7×7 blocks;
  - extrapolation: the top row ($c_k = 0.8$); the 51 or 75 outer runs, training on the centre;
  - four training runs, one step in from each corner (Section 5).
- Metric: RMSE over the water cells against the last-year mean, per held-out run, then the mean and median over the runs.
- Paired comparison with the GP on s, on the per-run difference (network RMSE averaged over the 4 initialisation sets, minus the GP's): its mean, a 95 % bootstrap interval, and a Wilcoxon signed-rank p-value. The p-values are also given with a Holm correction over the eight main evaluations (the four-run split excluded).
- Sensitivity (Table 3): the test RMSE of every configuration on every split, averaged over the 4 initialisation sets, and the choice of both rules when the cross-validation uses other partitions. Neither is used for the selection.
- Reproducibility: every fit runs with one BLAS thread.
  - With the same library versions (scikit-learn 1.7.2, NumPy 2.2.5, OpenBLAS 0.3.30), a refit reproduces every number bit for bit. The sensitivity check refits the chosen configurations in other processes and matches every per-run RMSE exactly.
  - With several threads, L-BFGS stops at a slightly different point. Compared with an earlier run on 8 threads, a split's mean moved by up to 0.0003 K, one initialisation set's mean by up to 0.0013 K, and a single run by up to 0.008 K. Differences of that size are not meaningful.
- Cost: a fit takes seconds on one core, and a prediction microseconds. On 8 cores the evaluation takes 46 minutes, mostly the repeated cross-validation and the 100 leave-one-out refits; the sensitivity check takes 28 minutes.

split	held out / train	network	range	median	GP on s	GP on z	RBF	nearest	mean state
scattered	10 / 90	**0.017**	0.0008	0.013	0.019	0.022	0.021	0.038	0.161
band (three rows)	30 / 70	**0.038**	0.0010	0.033	0.045	0.051	0.057	0.118	0.204
centre 5×5	25 / 75	**0.025**	0.0004	0.019	0.029	0.035	0.037	0.080	0.162
centre 7×7	49 / 51	**0.029**	0.0006	0.026	0.035	0.069	0.068	0.116	0.203
leave-one-out	100 / 99	**0.024**	0.0004	0.017	0.028	0.032	0.034	0.085	0.221
top row	10 / 90	**0.049**	0.0028	0.033	0.079	0.083	0.084	0.176	0.482
outer 51	51 / 49	**0.067**	0.0005	0.027	0.074	0.081	0.093	0.134	0.265
outer 75	75 / 25	**0.114**	0.0009	0.041	0.125	0.131	0.138	0.159	0.226

Table 1: RMSE (K) of the last-year temperature over the water cells, mean over the held-out runs. Network: mean over the 4 initialisation sets; range: how much that mean varies across them; median: over the runs. Bold: lowest mean per split. The four-run split is in Section 5.

	one-standard-error rule (reported)			minimum CV RMSE				
split	difference [95 % interval]	p	Holm p	choice	network	difference	p	Holm p
scattered	-0.0025 [-0.0053, -0.0005]	0.037	0.074	10^{-3}	0.016	-0.0030	0.049	0.195
band (three rows)	-0.0075 [-0.0132, -0.0020]	0.015	0.044	10^{-3}	0.037	-0.0083	0.011	0.068
centre 5×5	-0.0045 [-0.0071, -0.0019]	0.003	0.013	10^{-4}	0.026	-0.0032	0.022	0.110
centre 7×7	-0.0058 [-0.0096, -0.0021]	0.003	0.013	10^{-3}	0.035	$+0.0010$	0.745	1.000
leave-one-out	-0.0046 [-0.0086, -0.0015]	< 0.001	< 0.001	10^{-3}	0.023	-0.0052	< 0.001	< 0.001
top row	-0.0297 [-0.0768, $+0.0001$]	0.193	0.193	10^{-4}	0.058	-0.0204	1.000	1.000
outer 51	-0.0076 [-0.0123, -0.0032]	0.002	0.013	10^{-3}	0.067	-0.0072	0.009	0.066
outer 75	-0.0111 [-0.0170, -0.0059]	< 0.001	0.004	10^{-2}	0.122	-0.0032	0.594	1.000

Table 2: Network minus GP on s, per held-out run (K; negative = network better): the mean difference, its 95 % bootstrap interval, the Wilcoxon signed-rank p and the Holm-corrected p over the eight comparisons. Left: the network as reported (one-standard-error rule). Right: the network chosen by the minimum CV RMSE instead (ReLU, with the α given), and its mean RMSE.

	configurations within one standard error				choice, other partitions		
split	number	test RMSE	below GP on s	rank of the choice	all ten	one SE	minimum
scattered	5	0.016–0.017	5	3	0.016–0.034	same	same
band (three rows)	6	0.037–0.040	6	4	0.037–0.072	same	10^{-2}
centre 5×5	5	0.025–0.030	3	1	0.025–0.046	same	10^{-3}
centre 7×7	6	0.029–0.045	1	1	0.029–0.060	same	10^{-4}
top row	5	0.049–0.058	5	1	0.049–0.146	same	10^{-3}
outer 51	5	0.067–0.077	3	1	0.067–0.121	same	same
outer 75	6	0.114–0.144	3	1	0.114–0.165	same	same
four runs	6	0.117–0.187	2	5	0.098–0.187	same	same

Table 3: Sensitivity to the configuration. Left: the configurations whose CV RMSE is within one standard error of the best, the range of their test RMSE (mean over the 4 initialisation sets), how many beat the GP on s, and the rank of the rule's choice among them (1 = lowest test RMSE). All ten: the range over the whole grid. Right: the choice of each rule when the cross-validation repeats use other partitions ("same": unchanged).

4 Findings

- **In interpolation the network has the lowest error of all methods** (Tables 1 and 2, Figure 2).
  - Its mean error is 0.25–0.44 times the nearest run's.
  - Against the GP on s the margin is 0.0025–0.0075 K. After the Holm correction it is significant on band, both centre blocks and leave-one-out ($p \leq 0.044$). It is not significant on the scattered split ($p = 0.074$; 7 of 10 runs better).
- **In extrapolation it is better on outer 51 and outer 75** (Holm $p = 0.013$ and 0.004).

  - On the top row the gain (0.049 vs 0.079 K) is mostly the warm-corner run (0.16 vs 0.37 K), and the test is not significant ($p = 0.19$).
  - Without that run, the network gets 0.037 vs 0.046 K and wins 6 of 9.
- **The margin over the GP on s depends on the selection rule** (Table 2, right).
  - The minimum CV RMSE picks a weaker regularisation. That network is still lower than the GP on s on 7 of the 8 evaluations.
  - It ties on the 7×7 block (+0.0010 K, $p = 0.75$) and on outer 75 (−0.0032 K, $p = 0.59$).
  - After the Holm correction only leave-one-out remains significant; band and outer 51 come closest ($p = 0.068$ and 0.066).
  - Its error stays below the GP on z, the RBF and the nearest run on every evaluation.
  - The one-standard-error rule is the more stable choice. With other cross-validation partitions it keeps its configuration on all eight splits, while the minimum changes on four (Table 3).
  - But the rule was adopted after the first review had seen test errors of other configurations (Section 2), so the margins under it are not a blind test.
- **Initialisation matters little, the configuration more** (Tables 1 and 3).
  - Four initialisation sets move a split's mean by 0.0004–0.0010 K, and by 0.0028 K on the top row.
  - Five or six configurations per split are within one standard error of the best CV score. Their test errors span up to 0.03 K (outer 75: 0.114–0.144 K).
  - On scattered, band and the top row, all of them beat the GP on s. On the 5×5 block, outer 51 and outer 75, 3 of them do; on the 7×7 block, 1 of 6.
  - The rule's choice has the lowest test error of the set on five of the seven splits. It ranks third of 5 on scattered and fourth of 6 on band.
- **The error sits in the upper** 500 m (Figures 3 and 4).
  - In interpolation it is 0.014–0.055 K there and 0.003–0.016 K below 1000 m.
  - At the centre of the 7×7 block, the predicted anomaly (relative to the training mean) has correlation 0.98 with the true one and amplitude 1.05.
  - This and the other example values are those of Figure 4, which shows the first initialisation set.
- **The warm corner** ($c_k = 0.8$, $c_\epsilon = 0.0875$) is the hardest state (Figure 4).
  - One row from the training set (top row), the network reproduces its warming: correlation 0.99, amplitude 0.99, 0.157 K.
  - Three cells away (outer 75), it keeps a weak version of the pattern (correlation 0.62) with a third of the amplitude (0.33): a cold bias of −0.68 K and 1.052 K RMSE.
  - Averaged over the four initialisation sets, the network's error there is 1.06 K. Every method is between 1.06 and 1.45 K.
- **Part of the extrapolation error is the basis.** Projecting the held-out truth on the training principal directions already leaves 0.010–0.017 K in interpolation and 0.031–0.079 K in extrapolation. The outer-75 value (0.079 K of the network's 0.114 K) is mostly the corner run, where the floor is 0.85 K.
- **The spread over initialisations is not an uncertainty.** It is 0.003–0.009 K in interpolation and 0.008–0.011 K in extrapolation, while the error grows from 0.017 to 0.114 K.
- The comparison is between a 5-member ensemble and single baselines, and only the network's hyper-parameters are tuned by cross-validation.

5 A failure case: four training runs

- Training on $c_k \in \{0.0198, 0.504\} \times c_\epsilon \in \{0.139, 3.53\}$ leaves 96 runs to predict, with $K = 3$ principal directions (2 inside the cross-validation folds).
- Cross-validation cannot choose here.
  - With four runs the 5-fold split is leave-one-out, so the three repeats are identical.
  - The standard error (0.15 K) covers six configurations.
  - Both rules pick ReLU with $\alpha = 1$, which also has the minimum CV RMSE.
- With that choice the network gets 0.147 K (0.134–0.169 over the initialisation sets). The GP on s gets 0.120 K, the RBF 0.146 K, the nearest run 0.181 K.

- Over the ten configurations the network ranges from 0.10 to 0.19 K (Table 3). Among the six within one standard error, the choice ranks fifth. The outcome is set by a choice that four runs cannot inform.
- Regimes:
  - where the state saturates ($s < -2$, 46 runs), the medians are 0.040 (nearest run), 0.058 (network) and 0.061 K (GP on s);
  - on the warm side (50 runs) they are 0.197 (network), 0.172 (GP on s) and 0.223 K (nearest run).
- With $\alpha = 1$ the network is too flat (Figure 4, rows 2 and 5; first initialisation set).
  - At the centre run its anomaly has the right pattern (correlation 0.98) with 45 % of the amplitude: 0.114 K (0.124 K over the four sets), against 0.036 K for the GP on s.
  - At the corner: correlation 0.86, amplitude 0.40, 0.801 K (0.831 K over the four sets).

6 Next steps

- Inputs $(s, s_\perp)$ instead of $(\log c_k, \log c_\epsilon)$, to build in the collapse that makes the GP on s strong with few runs.
- For extrapolation, either a decoder not restricted to the span of the training states, or more runs near the warm corner.
- An uncertainty from resampling the training runs rather than the initialisations.
- Tune the baselines by cross-validation as well (RBF smoothing, GP noise and mean function), for an even comparison.
- Fix the selection rule, or α itself, in advance, and test it on new runs or new splits. Here the rule was revised after test errors had been seen, and with very few runs cross-validation cannot choose at all.

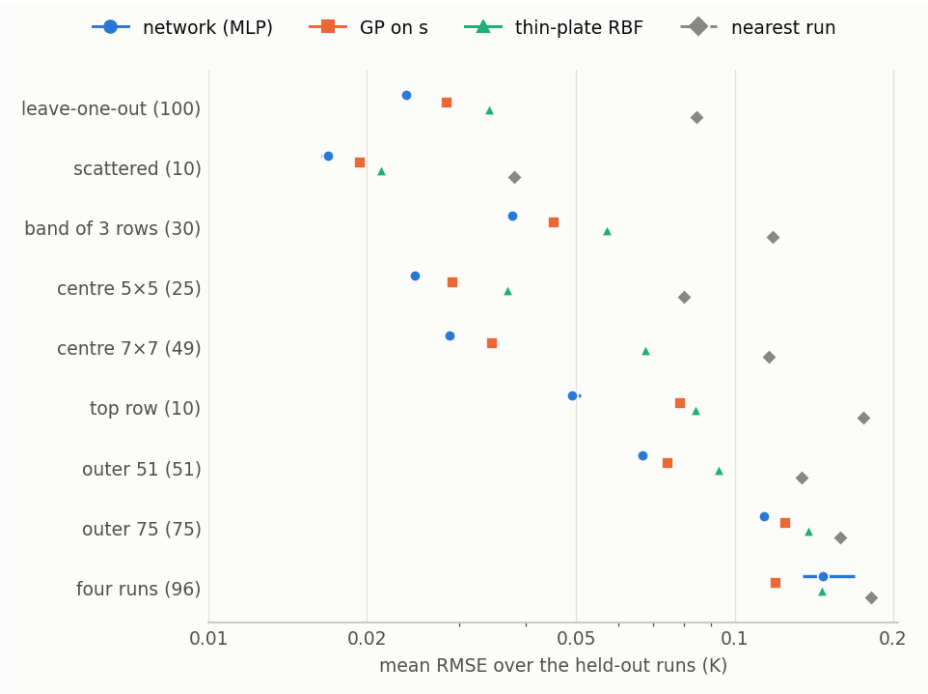


Figure 2: Mean RMSE over the held-out runs per split (log scale) for the network and three baselines: the GP on s, the thin-plate RBF and the nearest run. Held-out run counts are in brackets. Network: mean over 4 initialisation sets; the bar is their range.

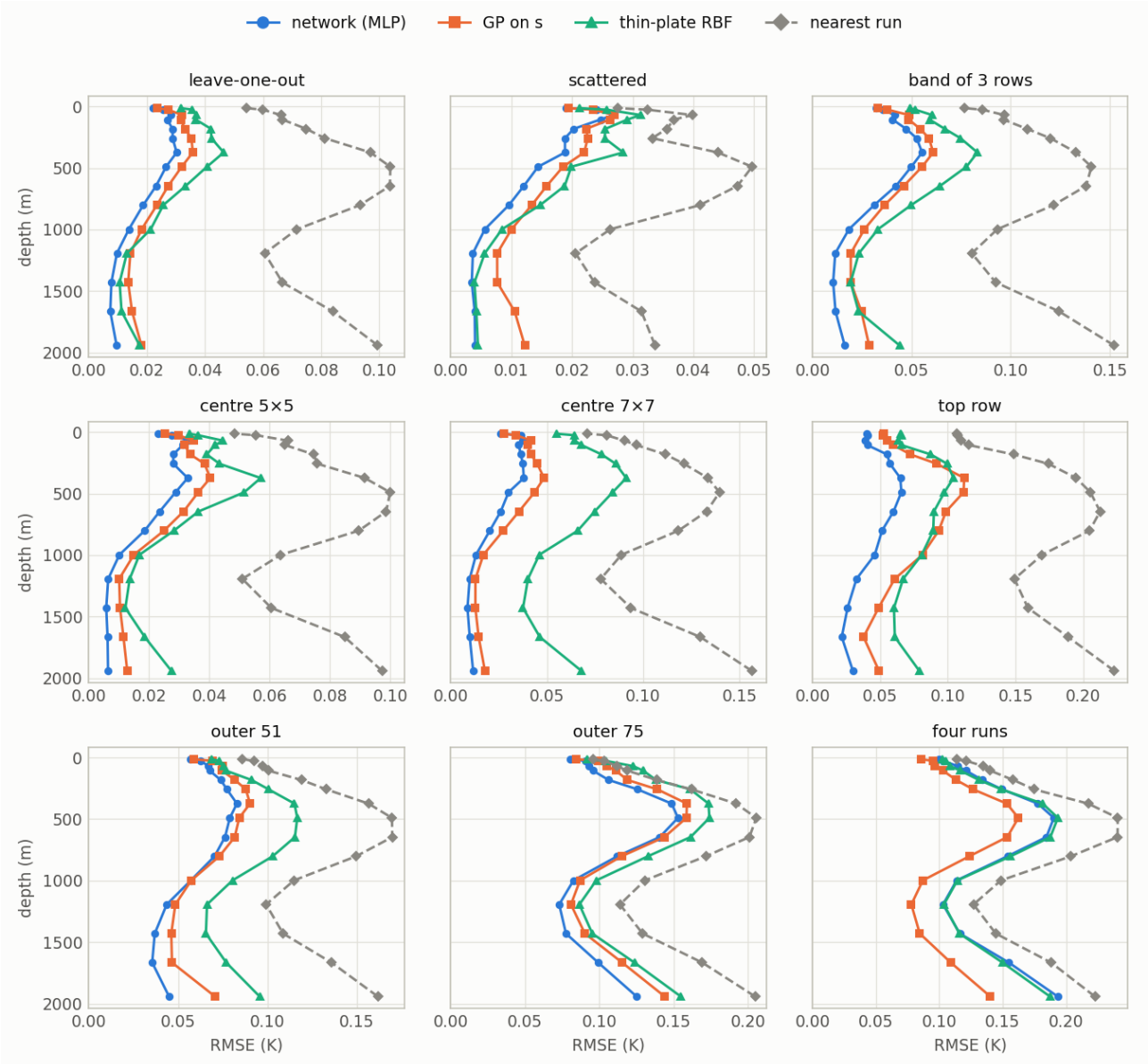


Figure 3: RMSE by depth, mean over the held-out runs of each split (network: averaged over the initialisation sets).

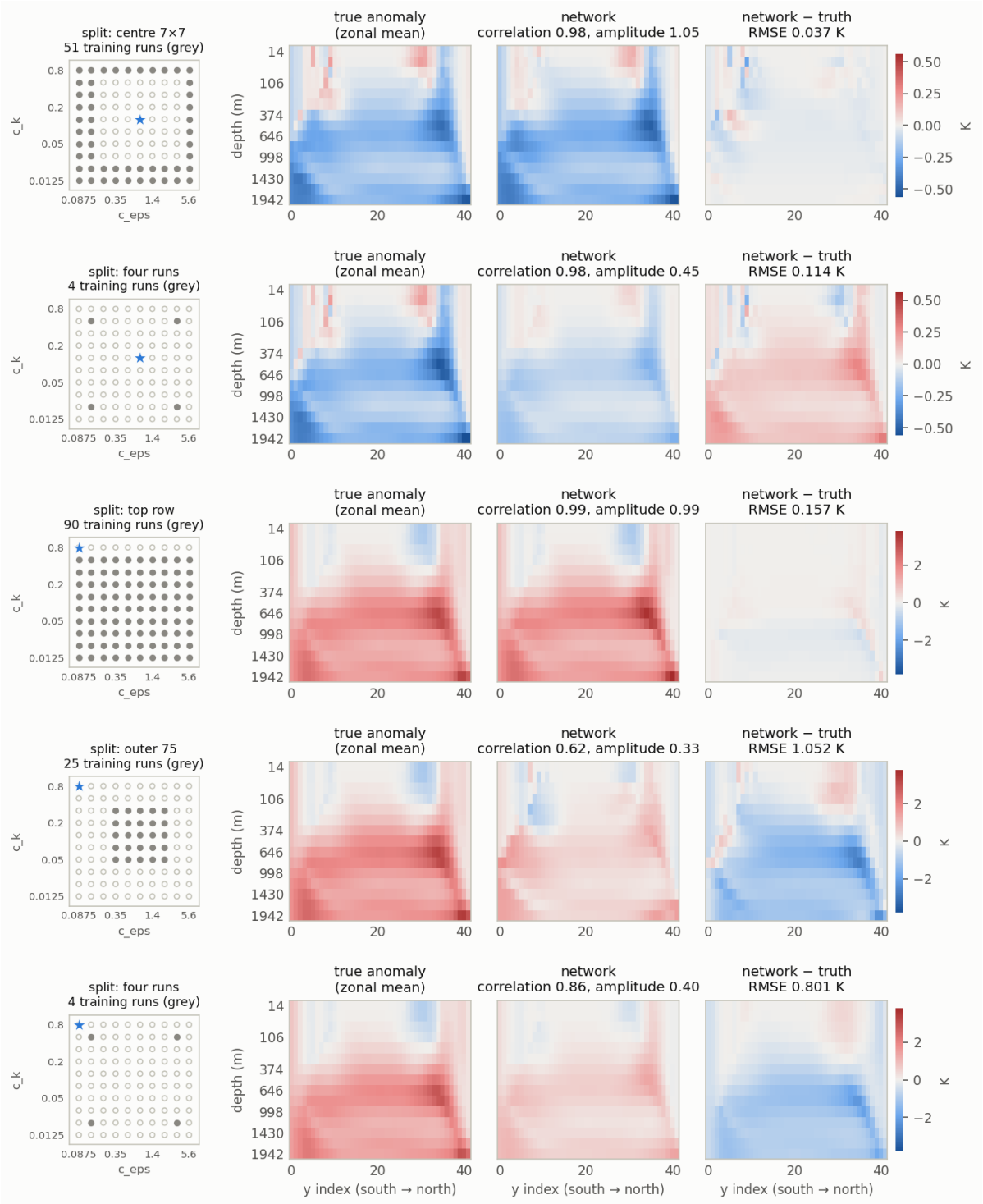


Figure 4: Zonal-mean sections of the last-year anomaly, relative to the mean of the split's training runs (what the network has to add to it): truth, network (first initialisation set) and their difference. Each run has one colour scale, and levels are drawn with equal height. Left: the split (grey: training runs; open circles: held out; star: the run shown). Rows 1–2: the centre run of the 7×7 block, held out with the block (51 training runs) and in the four-run split. Rows 3–5: the warm-corner run, held out with the top row (90 training runs), with the outer 75 (25) and in the four-run split (4). Correlation and amplitude (the regression slope of the predicted on the true anomaly) are over all water cells.